

Mirror Descent

Revisiting Gradient Guarantees — dimension independent

Recall: $\min_x f(x)$
 $s.t.: x \in \mathcal{X}$

$$x_{t+1} = \text{Proj}_{\mathcal{X}}(x_t - \eta g_t), \quad g_t \in \partial f(x_t)$$

Thm: If f is L -Lipschitz, taking

$$\eta \sim \frac{1}{L} \cdot \frac{1}{\sqrt{T}}$$

Then: $f\left(\frac{1}{T} \sum_{t=1}^T x_t\right) - f(x^*) \leq L \cdot \left(\frac{1}{\sqrt{T}}\right)$

Sub gradient method: linear approximation + proximity

$$\hookrightarrow \boxed{x_{t+1} = x_t - \eta g_t}$$

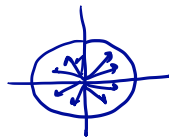
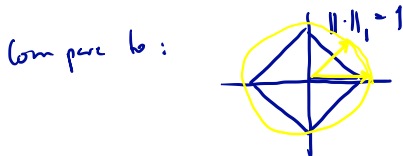
min: $f(x)$

$$x_{t+1} = \underset{x}{\operatorname{argmin}}: \underbrace{f(x_t) + g_t^T(x - x_t)}_{\text{lin. approx to } f \text{ at } x_t} + \underbrace{\frac{1}{2\eta} \|x - x_t\|_2^2}_{\text{prox term}}$$

$$\approx \underset{x}{\operatorname{argmin}}: \underbrace{\eta g_t^T x}_{\text{lin. approx}} + \underbrace{\frac{1}{2} \|x - x_t\|_2^2}_{\text{prox term}}$$

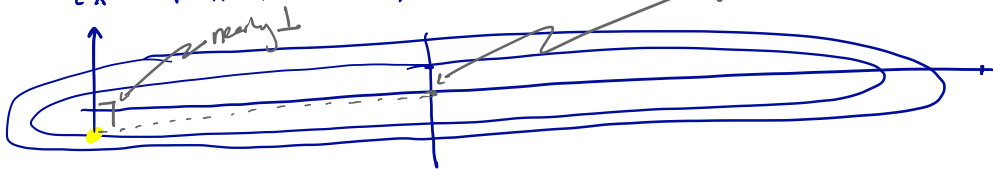
prox term is Euclidean norm

Euclidean geometry: spherical symmetry



An example - part 1 - the picture

$$E_X: f(x_1, x_2) = x_1^2 \cdot \frac{1}{100} + x_2^2 \cdot 100$$



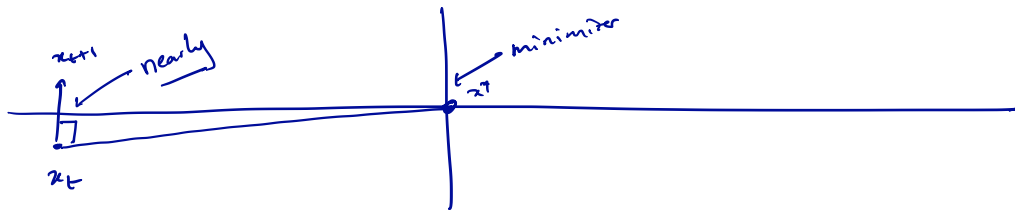
suppose we are at the point: $x_t = \begin{pmatrix} -10 \\ -0.1 \end{pmatrix}$

sub-gradient method: $x_t \mapsto x_{t+1} = x_t - \gamma \nabla f(x_t)$

$$\nabla f(x_t) = \begin{pmatrix} 2x_1/100 \\ 2x_2 \cdot 100 \end{pmatrix} \bigg|_{\begin{pmatrix} -10 \\ -0.1 \end{pmatrix}} = \begin{pmatrix} -1/5 \\ -20 \end{pmatrix} \quad \text{move in } - \begin{pmatrix} -1/5 \\ -20 \end{pmatrix}$$

An example - part 2 - the problem

b/c of the elongation of the level sets
the direction of movement: $(x_{t+1} - x_t) \perp (x^* - x_t)$



An example - part 3 - a solution?

Sub-g. method: $x_{t+1} = \underbrace{\eta \nabla f(x_t)^T x_t}_{\text{argmin lin approx at } x_t} + \frac{1}{2} \|x - x_t\|_2^2 \quad \frac{1}{2} (x - x_t)^T \uparrow (x - x_t)$

Instead: $x_{t+1} = \underset{x}{\text{argmin}}: \eta \nabla f(x_t)^T x + \frac{1}{2} (x - x_t)^T \begin{pmatrix} \frac{1}{50} & 0 \\ 0 & \frac{1}{200} \end{pmatrix} (x - x_t)$

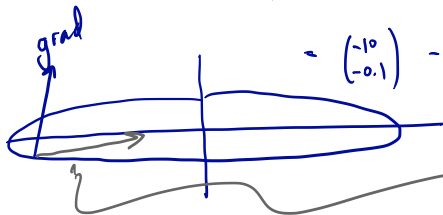
Take deriv, set $= 0$, and find x_{t+1} in this new alg.

$$\eta \nabla f(x_t) + \begin{pmatrix} \frac{1}{50} & 0 \\ 0 & \frac{1}{200} \end{pmatrix} (x - x_t) = 0$$

solve for x :

$$x_{t+1} = x_t - \eta \begin{pmatrix} 50 & 0 \\ 0 & \frac{1}{200} \end{pmatrix} \nabla f(x_t)$$

$$= \begin{pmatrix} -10 \\ -0.1 \end{pmatrix} - \eta \begin{pmatrix} 50 & 0 \\ 0 & \frac{1}{200} \end{pmatrix} \begin{pmatrix} -1/5 \\ -20 \end{pmatrix} = x_t - \eta \begin{pmatrix} -10 \\ -0.1 \end{pmatrix}$$



new update

new direction
↓

Related to Newton's method.

An example - part 4 - Lipschitzness

We changed the prox term.

$\Rightarrow$ This changed the direction: $(x_{t+1} - x_t)$

If we measure distance using this new way, then we also change its Lipschitzness -

Using Euclidean norm: $|f(x) - f(y)| \leq L \cdot \underbrace{\|x - y\|_2}_{\text{Euclidean}}$

What if instead: $\|z\|^2 = z^T \begin{pmatrix} 1/50 & 0 \\ 0 & 200 \end{pmatrix} z$

Exercise: • What is the Lipschitz constant of f using Euclidean norm, at the point $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

• What about the new norm?

An example - part 5 - observations

Observation 1: $x_{t+1} = \arg\min: \eta \cdot g_t^T x + \frac{1}{2} \|x - x_t\|_L^2$

- Changing the proximity term to a different norm gives a different algorithm. May be better.

Observation 2: Different Lipschitz constant using different norm.

Connection: Our guarantees for convergence rate have L in there. So far we have focused only on dependence on T .